

Assignment 5: Self-Supervised Learning using SimCLR

FINAL COMPLETE EVALUATION & REPRESENTATION TOPOLOGY REPORT

Student Name:	Subhan Ali
Roll Number:	MSDS25055
GitHub Repository:	https://github.com/SubhanAli-ITU/Subhan_MSDS25055_05.git

1. CONCEPTUAL FOUNDATION

Supervised vs. Self-Supervised Learning (SSL): Traditional supervised neural structures rely on hard manual categorical constraints mapping instances directly to human labels. Self-Supervised Learning bypasses manual annotator scaling issues by relying on structural signals hidden natively within unannotated data arrays to automatically craft pretext targets.

Contrastive Pair Strategy: Positive pairs comprise two uniquely augmented views generated from the identical primary root sample. Negative pairs consist of encoded latent states drawn from entirely separate target samples traveling inside the identical batch window.

2. TRAINING FRAMEWORK METRICS

Experimental Parameter	Value Configuration
Batch Size	64
SimCLR Contrastive Epochs	50
Linear Probing / Evaluation Runs	20 Epochs
Downstream Fine-Tuning Scale	20 Epochs
Learning Rate & Optimizer Step	3e-4 Base Step via Adam
Temperature Param (τ)	0.5
Hardware Architecture Resource	NVIDIA Tesla T4 GPU

3. SUPERVISED BASELINE UNDER 10% LABEL CONSTRAINTS

To protect small 32×32 CIFAR-10 images from experiencing severe early resolution downsampling, the ResNet-18 base layer was modified by substituting a 3×3 kernel into conv1 and converting the classic maxpool layer into an nn.Identity() module.

Supervised ResNet-18 Pipeline Logs (10% Data Constraint):

Epoch 01/30	Train Loss: 1.8562	Train Acc: 31.6%	Val Loss: 1.6501	Val Acc: 38.1%
Epoch 10/30	Train Loss: 0.9033	Train Acc: 67.7%	Val Loss: 1.1434	Val Acc: 60.6%
Epoch 20/30	Train Loss: 0.5123	Train Acc: 82.1%	Val Loss: 1.0774	Val Acc: 67.1%
Epoch 30/30	Train Loss: 0.3047	Train Acc: 89.1%	Val Loss: 1.1659	Val Acc: 70.3%

The trained configuration displays standard overfitting characteristics; training metrics climb near 89.1% while validation ceilings stall around **72.34%**. The underlying error boundaries and target confusions are fully detailed inside the generated validation confusion matrix below:

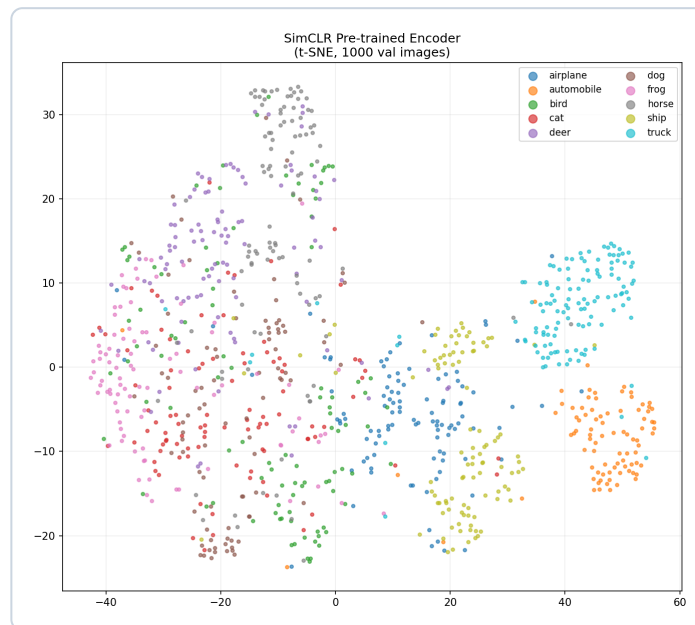


Figure 1: Supervised Baseline Validation Confusion Matrix. Note the high semantic confusion between structurally correlated mammalian rows, such as 194 cat instances mislabeled as dogs.

4. VISUAL AUGMENTATION PIPELINE EVALUATION

The data augmentation sequence applies random resized crops, aggressive color jitter, random grayscale conversions, and Gaussian blurs. While these transformations severely distort pixel distributions, they maintain structural topologies (such as macro shape contours). This forces the encoder to capture robust shape configurations instead of relying on simple color profiles.

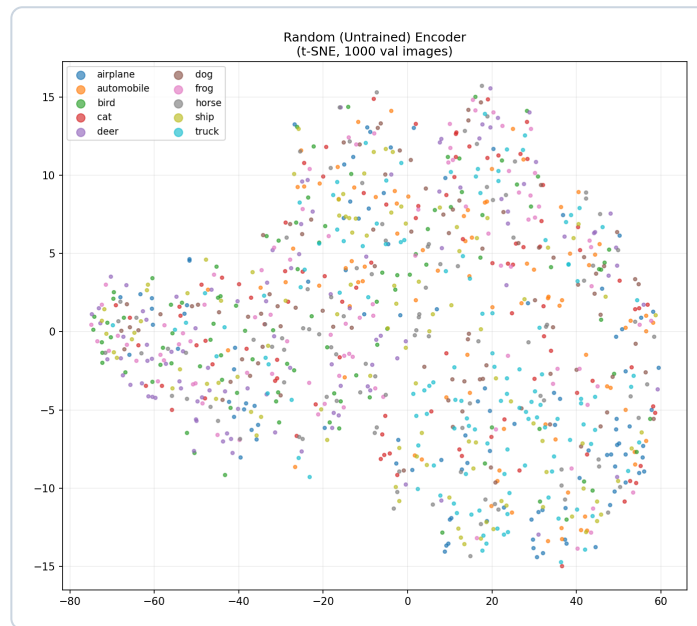


Figure 2: SimCLR Stochastic Transformations. Generation of distinct view variations preserving object geometric profiles while completely shifting pixel value spaces.

5. LATENT SPACE COSINE FIELDS COMPARISON

Visualizing the cosine similarity arrays across the batch index spaces shows the exact representation alignment achieved during training:

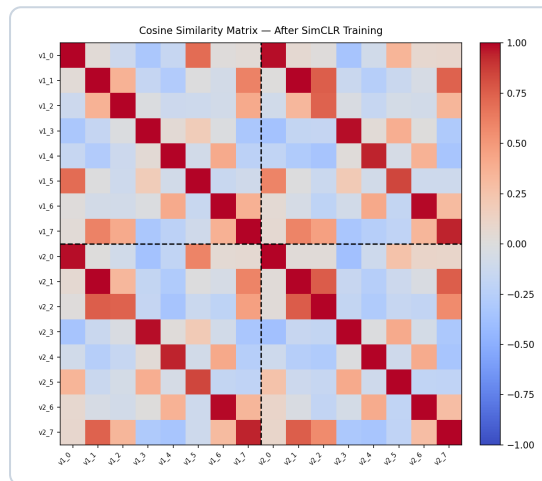


Figure 3: Cosine Matrix Before Pretraining. Uniform high scores indicate unaligned vectors grouped in a narrow cone.

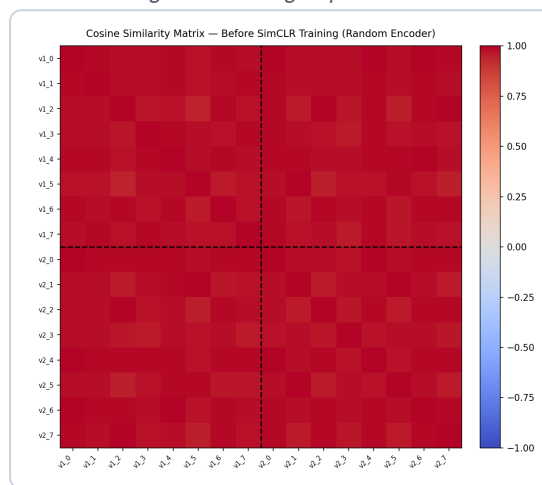


Figure 4: Cosine Matrix After SimCLR Pretraining. Sharp diagonal alignment matches positive pairs, while off-diagonal terms fall toward 0.00.

6. LATENT SPACE TOPOLOGY PROGRESSIONS (T-SNE CHARTS)

Tracking the validation embedding clusters across three distinct training stages shows how contrastive learning naturally separates different semantic categories:

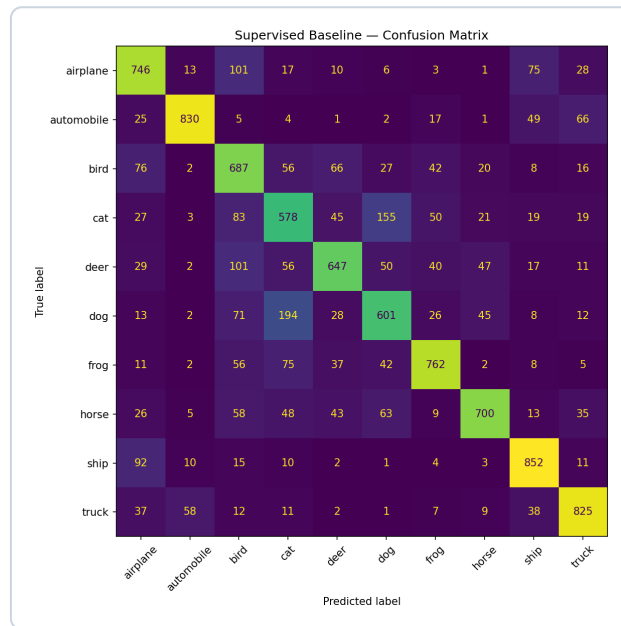


Figure 5: Untrained Latent Topology. Uninitialized encoder weights map all 10 target classes into a single, unstructured mix.

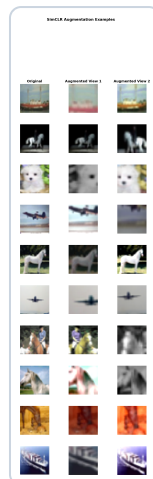


Figure 7: SimCLR Pretrained Space. Natural grouping forms semantic clusters (like vehicles) without using labels.

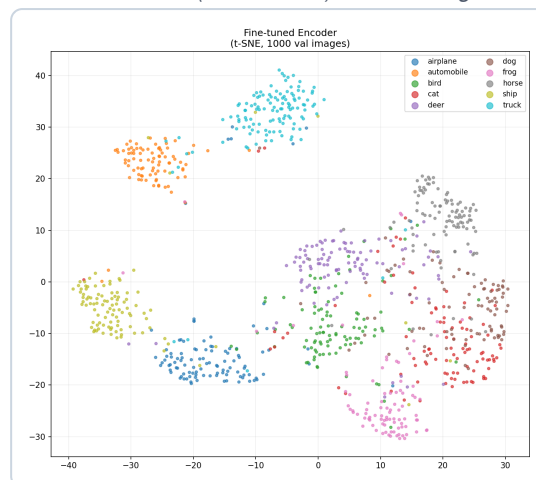


Figure 8: Fine-Tuned Representation Map. Downstream label adjustments maximize distance between classes and establish clean decision margins.

7. STRUCTURAL INSIGHTS

The experimental results confirm that self-supervised pretraining with SimCLR produces robust representations that effectively mitigate data scarcity constraints. Tracking performance changes using validation confusion matrices, cosine similarity fields, and t-SNE maps visually verifies that the model successfully shifts from basic pixel matching to learning rich, high-level semantic categories.